

Jensen vs IVV: performance and risk attribution

Window: holdings 2019-09-30 to 2026-06-30, returns 2019-09-30 to 2026-09-30.

Annualised book return over the 28 quarters: the fund 10.65%, IVV 16.27%. Cumulative excess return D: -84.19 percentage points.

Data: mean unmapped weight of the fund book 0.07% and mean unpriced weight 0.01% over the 28 quarters.

Table 1. Brinson-Fachler effects over the 28 quarters, linked with Carino, in percentage points.

Bucket	Allocation	Selection	Interaction	Total
NoDur	-3.51	-1.47	-1.34	-6.31
Durbl	2.22	-2.59	2.48	2.12
Manuf	-3.54	-15.21	3.88	-14.87
Enrgy	0.16	0.00	0.00	0.16
Chems	-1.11	-0.27	-0.11	-1.50
BusEq	5.59	-29.36	-4.77	-28.54
Telcm	4.63	0.00	0.00	4.63
Utils	4.22	0.00	0.00	4.22
Shops	-2.32	-10.62	4.35	-8.58
HLth	-2.52	-10.54	-11.07	-24.13
Money	6.49	4.16	-8.35	2.31
Other	-7.94	-1.81	-3.90	-13.65
Unmapped	0.00	-0.19	0.14	-0.05
Unpriced	0.00	-0.10	0.10	0.00
Total	2.39	-67.99	-18.60	-84.19

Mean quarterly allocation 0.03% (90% interval -0.26% to 0.32%); mean quarterly selection -1.10% (90% interval -2.32% to 0.11%). Stationary bootstrap.

Jensen vs IVV: cumulative allocation and selection (Carino)

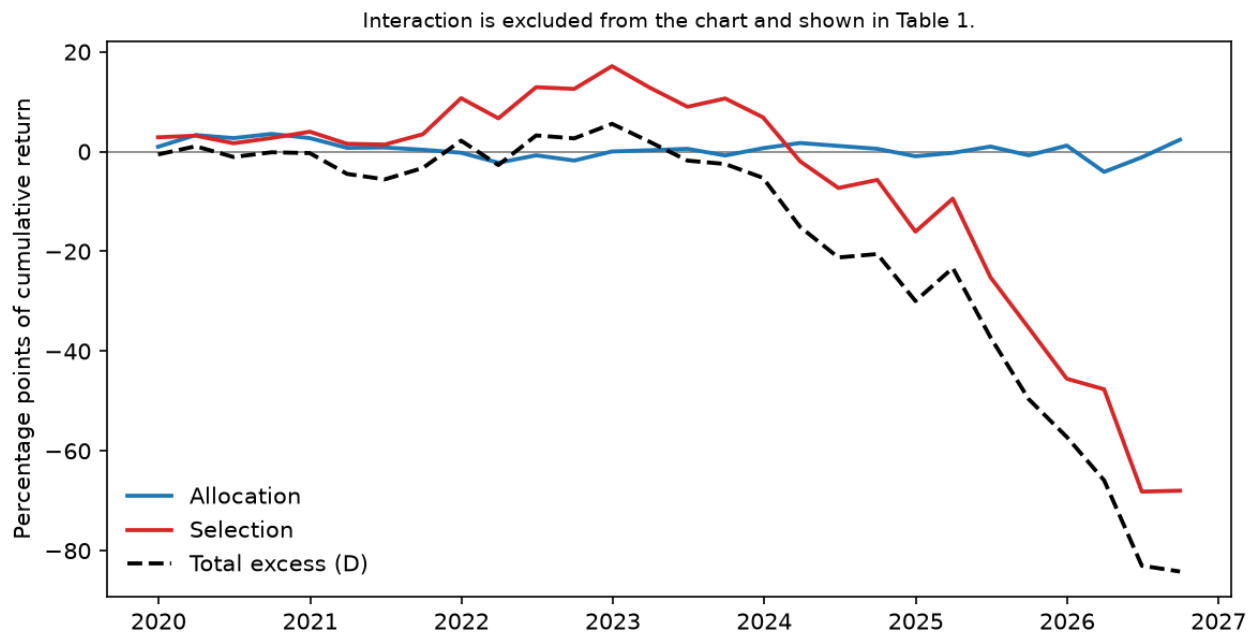


Table 2. Factor attribution of the fund book's monthly excess return by calendar year, with full-sample betas, in percent.

Year	Months	Excess	Mkt-RF	SMB	HML	RMW	CMA	UMD	Alpha	Residual
2019	3	7.89	7.69	-0.17	0.16	-0.33	-0.12	0.01	-0.96	1.60
2020	12	19.59	21.94	-0.68	2.61	-0.59	-1.12	0.00	-3.83	1.26
2021	12	27.99	19.51	0.10	-1.60	6.41	1.25	0.01	-3.83	6.13
2022	12	-16.65	-18.44	0.12	-2.21	2.42	3.36	-0.05	-3.83	1.98
2023	12	12.70	17.80	0.33	0.80	1.15	-2.14	0.06	-3.83	-1.47
2024	12	6.04	15.91	1.06	0.50	1.18	-1.08	-0.05	-3.83	-7.67
2025	12	2.02	11.26	0.81	-0.52	-2.92	-0.58	0.01	-3.83	-2.21
2026	8	3.89	9.20	-0.63	-0.62	-2.50	0.65	-0.02	-2.55	0.36

Full sample of 83 months: alpha -0.32% a month, HAC t -2.23, R² 0.92.

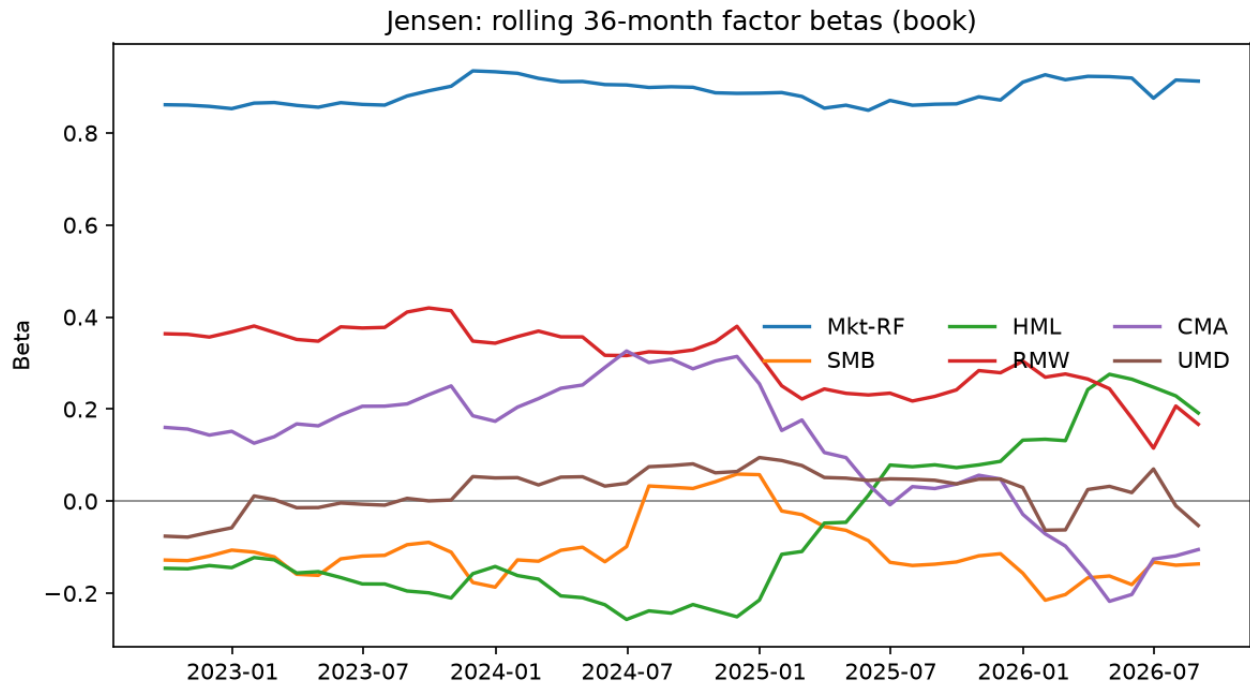


Table 3. Active share and ex-ante tracking error at each year end and at the last holdings date, in percent.

Date	Active share	Ex-ante TE	Names	Ridged
2019-12-31	75.89	3.36	461	No
2020-12-31	74.00	5.27	469	No
2021-12-31	71.46	5.59	481	No
2022-12-31	72.94	5.66	493	No
2023-12-31	72.15	5.14	495	No
2024-12-31	71.73	6.43	501	No
2025-12-31	58.82	4.76	510	No
2026-06-30	61.57	5.75	513	No

Realised tracking error over 84 months: 5.26%. Mean ex-ante tracking error over the 28 holdings dates: 5.27%.

Jensen vs IVV: top 15 contributions to ex-ante tracking error, 2026-06-30

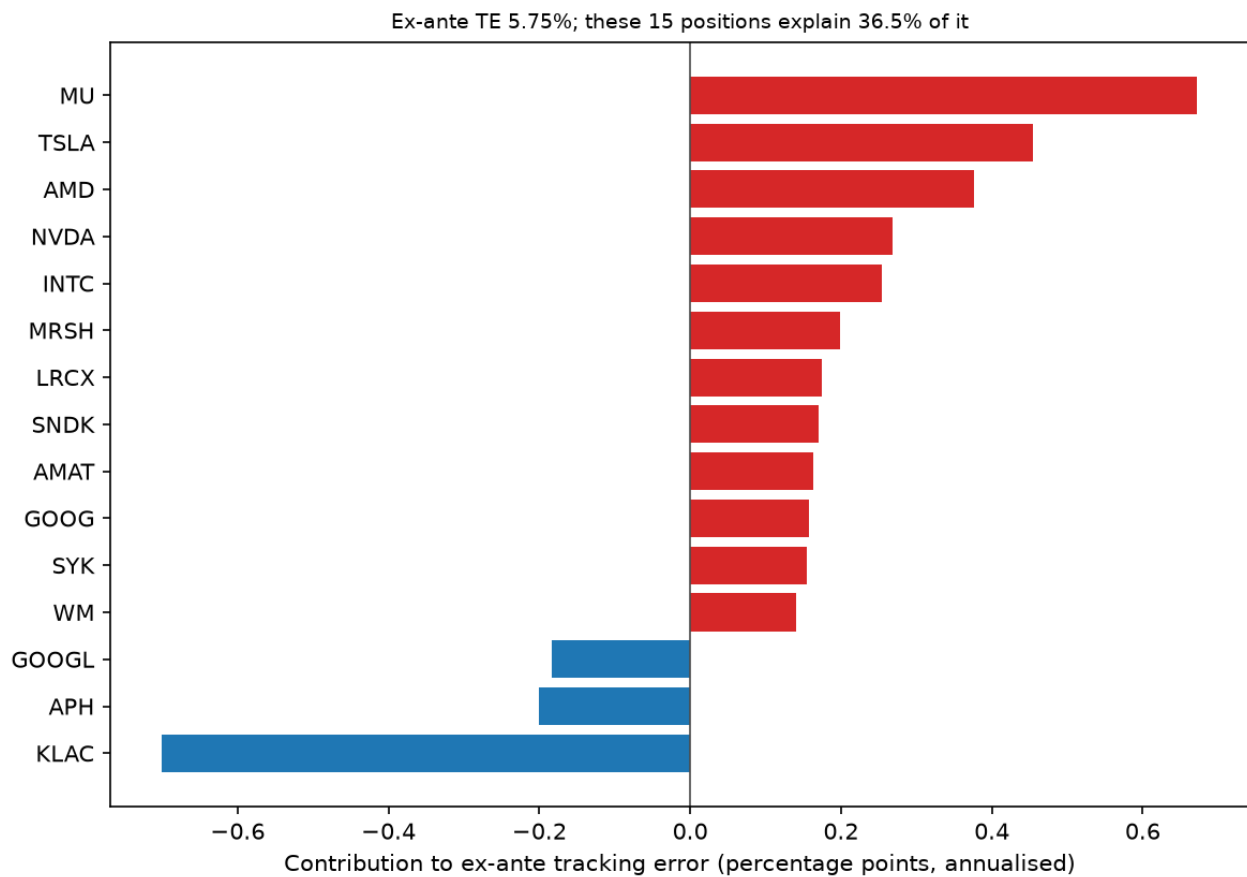


Table 4. Quarterly gap between the 13F book return and the fund's NAV return, in percent.

<u>Correlation</u>	<u>Quarters</u>	<u>Mean gap</u>	<u>Std of gap</u>	<u>TE of gap</u>	<u>90% interval of mean gap</u>
1.00	28	0.21	0.29	0.59	0.14 to 0.29

What a 13F book cannot see. The 13F lists a manager's long US-listed equity positions at each quarter end, filed up to 45 days later. It omits cash, shorts, most non-US shares, bonds and anything bought and sold inside the quarter. This report holds each quarter-end book unchanged for the next 3 months, so trades made during the quarter show up only in the gap between the book and the fund's NAV, together with fees and cash. A 13F belongs to the manager, not to one fund, so where a manager runs several strategies the book blends them. Sectors are Fama-French 12 industries built from SIC codes, which put payment networks such as Visa and Mastercard in Other. Securities that could not be mapped to a ticker, or had no price, earn the book's own return so that they move nothing.